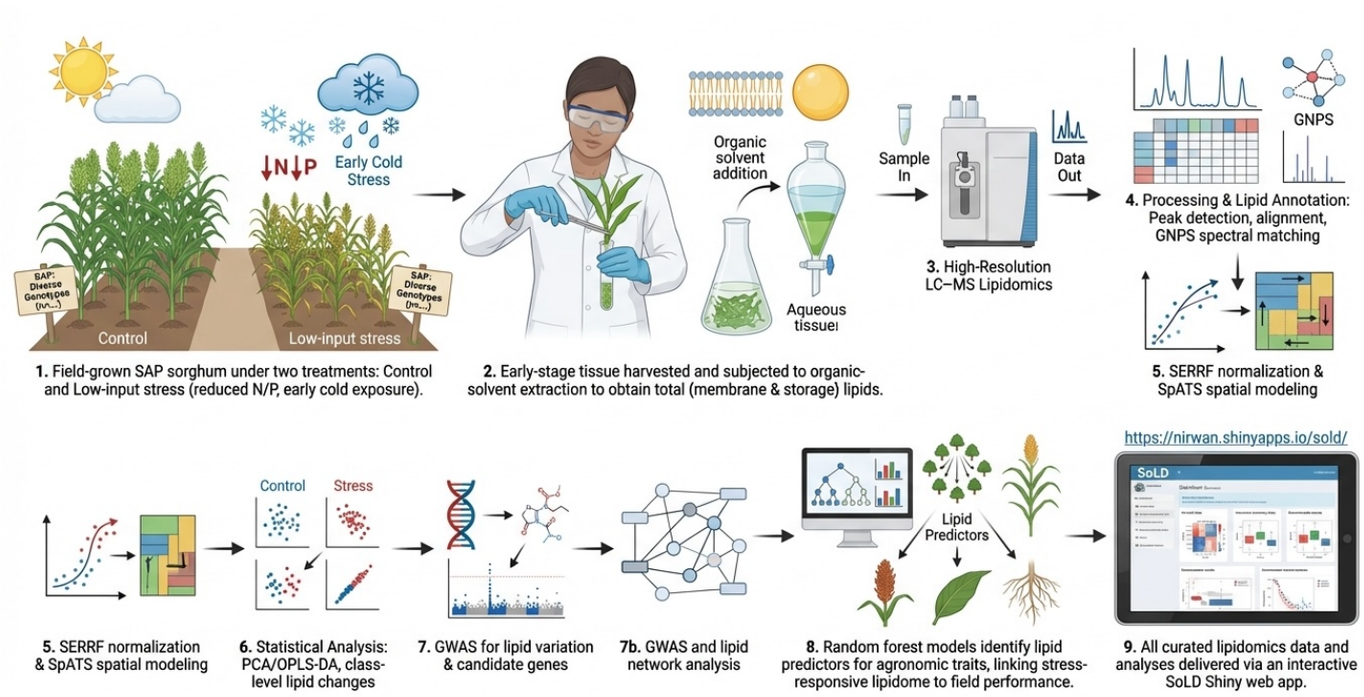
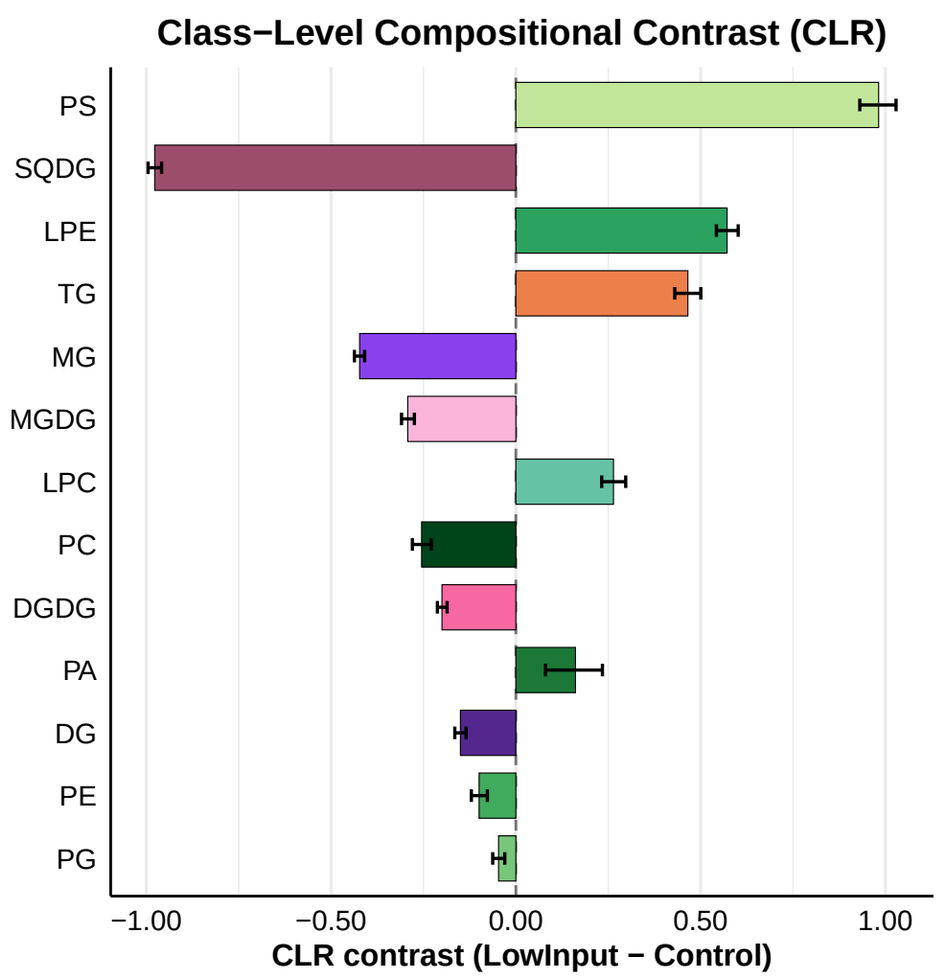
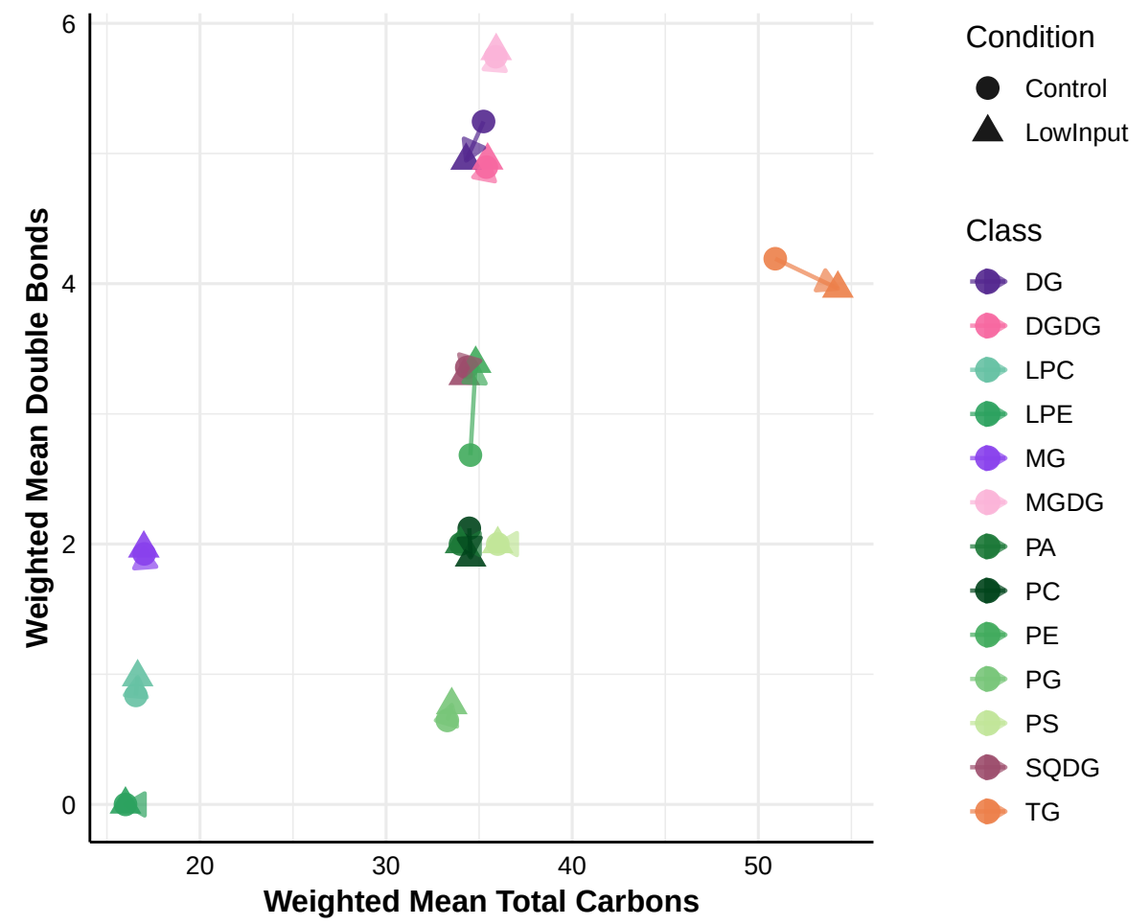
A


Early Cold Stress
↓N↓P
Control
Low-input stress
1. Field-grown SAP sorghum under two treatments: Control and Low-input stress (reduced N/P, early cold exposure).
Organic solvent addition
Aqueous tissue
2. Early-stage tissue harvested and subjected to organic-solvent extraction to obtain total (membrane & storage) lipids.
Sample In
Data Out
3. High-Resolution LC–MS Lipidomics
GNPS
4. Processing & Lipid Annotation: Peak detection, alignment, GNPS spectral matching
5. SERRF normalization & SpATS spatial modeling
https://nirwan.shinyapps.io/sold/
Control
Stress
Lipid Predictors
SoLD
5. SERRF normalization & SpATS spatial modeling
6. Statistical Analysis: PCA/OPLS-DA, class-level lipid changes
7. GWAS for lipid variation & candidate genes
7b. GWAS and lipid network analysis
8. Random forest models identify lipid predictors for agronomic traits, linking stress-responsive lipidome to field performance.
9. All curated lipidomics data and analyses delivered via an interactive SoLD Shiny web app.

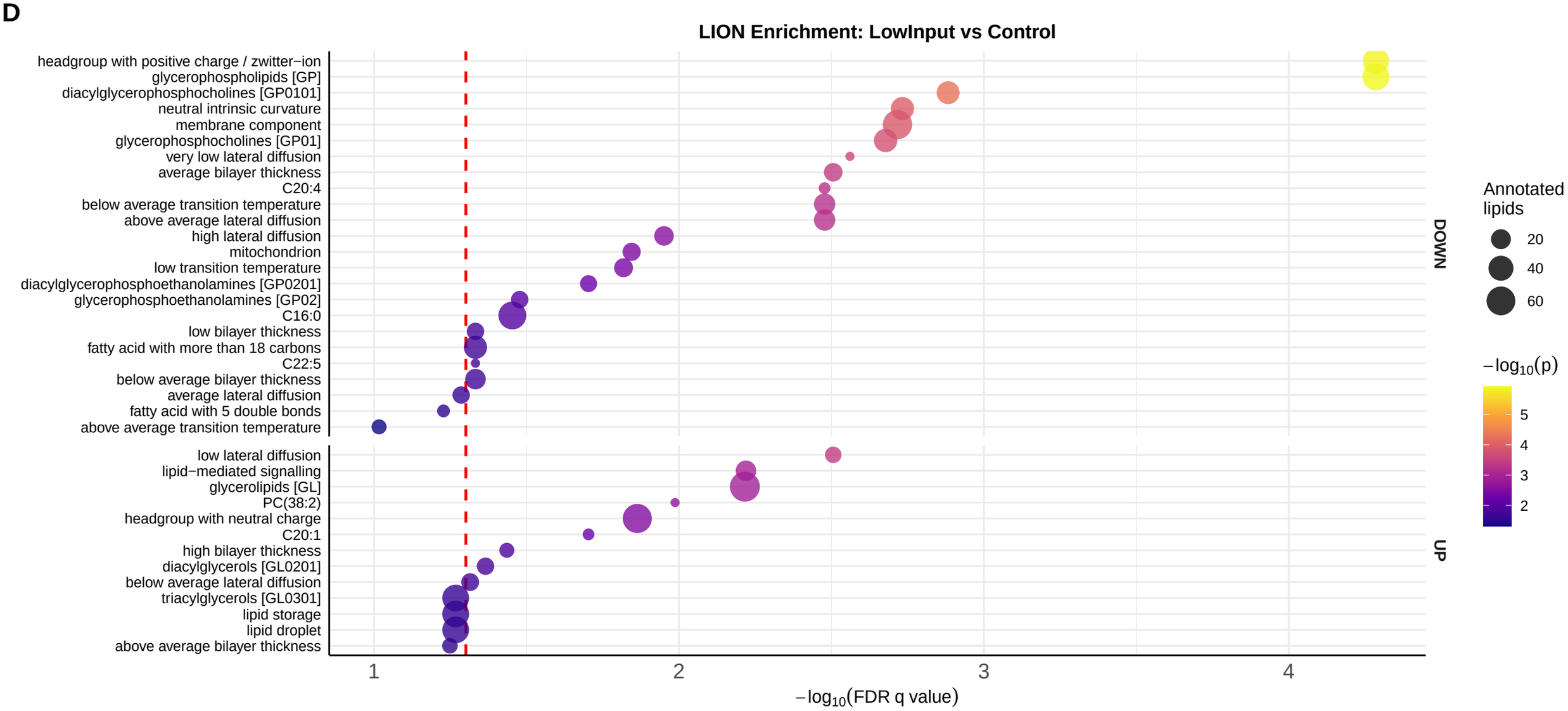
B


Class−Level Compositional Contrast (CLR)
PS
SQDG
LPE
TG
MG
MGDG
LPC
PC
DGDG
PA
DG
PE
PG
−1.00
−0.50
0.00
0.50
1.00
CLR contrast (LowInput − Control)

C


Chemical Shifts Between Conditions
Weighted Mean Double Bonds
0
2
4
6
20
30
40
50
Weighted Mean Total Carbons
Condition
Control
LowInput
Class
DG
DGDG
LPC
LPE
MG
MGDG
PA
PC
PE
PG
PS
SQDG
TG

D


LION Enrichment: LowInput vs Control
headgroup with positive charge / zwitter−ion
glycerophospholipids [GP]
diacylglycerophosphocholines [GP0101]
neutral intrinsic curvature
membrane component
glycerophosphocholines [GP01]
very low lateral diffusion
average bilayer thickness
C20:4
below average transition temperature
above average lateral diffusion
high lateral diffusion
mitochondrion
low transition temperature
diacylglycerophosphoethanolamines [GP0201]
glycerophosphoethanolamines [GP02]
C16:0
low bilayer thickness
fatty acid with more than 18 carbons
C22:5
below average bilayer thickness
average lateral diffusion
fatty acid with 5 double bonds
above average transition temperature
DOWN
low lateral diffusion
lipid−mediated signalling
glycerolipids [GL]
PC(38:2)
headgroup with neutral charge
C20:1
high bilayer thickness
diacylglycerols [GL0201]
below average lateral diffusion
triacylglycerols [GL0301]
lipid storage
lipid droplet
above average bilayer thickness
UP
1
2
3
4
−log10(FDR q value)
Annotated lipids
20
40
60
−log10(p)
5
4
3
2